

Senior Thesis

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```
rhfs <- read.csv("RHFS_2024_Public_Variables.csv")

df <- rhfs |>
  select(-REPWGT0) |> # REPWGT0 is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was done in Census's Methodol
    mse = TRUE
  )

# Sanity check against Table 5 of Census's Methodology
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
  )

## # A tibble: 1 x 3
##   properties properties_se   moe
##   <dbl>         <dbl> <dbl>
## 1    18965.         165.  271.

# No explicit NAs
sum(is.na(df))

## [1] 0
```